Quiz 5

1) Linear production function : $Y = a \cdot N$.

Recall that a concave production function is : $Y = a \cdot N^{\alpha}$

Hence, a linear production function is the special case when $\alpha = 1$. For a concave production function, α should be between 0 and 1.

Now, we need to analyze the labor demand when $\alpha = 1$:

$$L^d(\theta, a) = \left[\frac{\alpha \cdot a}{w \cdot (1+\tau(\theta))^{\alpha}}\right]^{\frac{1}{1-\alpha}}$$

$$\left(L^d(\theta, a)\right)^{1-1} = \left[\frac{1 \cdot a}{w\,(1+\tau(\theta))^{1}}\right]$$

$$1 = \frac{a}{w\,(1+\tau(\theta))}$$

$$a = w\,(1+\tau(\theta)) \quad (1)$$

Recall that, for a linear production function, MPL = a $\left(\frac{\partial Y}{\partial N} = a\right)$

Recall the expression for bargained wage (from the Wage functions lectures)

$$w = (1-\beta)z + z \cdot MPL\,(1 + r \cdot \theta) \quad (2)$$

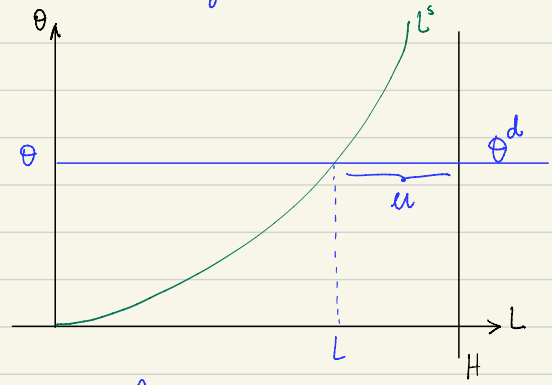
Plug (2) into (1): MPL = a

$$a = (1+\tau(\theta))\left[(1-\beta)z + z \cdot a\,(1+r\theta)\right] \quad \div a \text{ (divide both sides by "a")}$$

$$1 = (1+\tau(\theta))\left[(1-\beta)\frac{z}{a} + z\,(1+r\theta)\right]$$

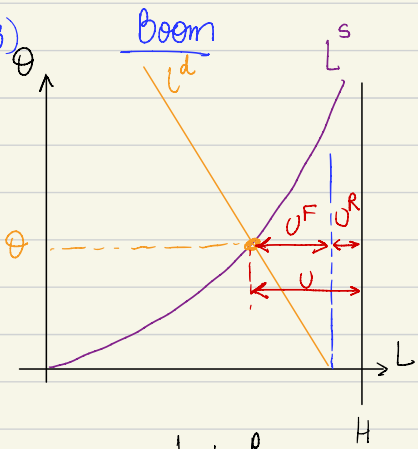
, which results in a perfectly elastic labor demand (horizontal $\theta^d(a)$)

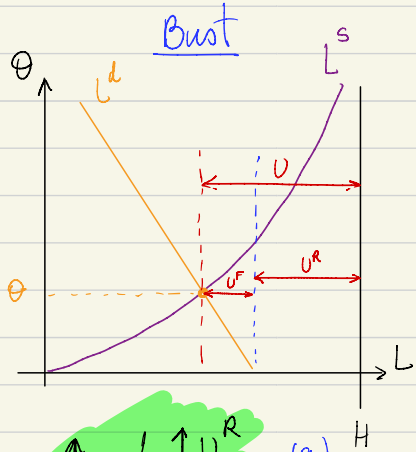
2) With a horizontal labor demand ($\theta^d(a)$):



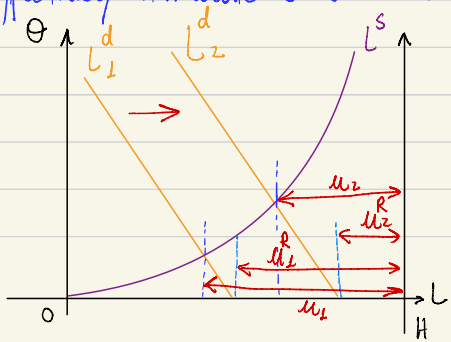
No, the job rationing model requires a concave production function, which allows for a downward-sloping $L^d(\theta, a)$

3)





$\downarrow U \begin{cases} \downarrow U^R \\ \uparrow U^F \end{cases}$

$\uparrow U \begin{cases} \uparrow U^R \\ \downarrow U^F \end{cases}$ (a)

4) In bad times, "U^R" is high. It happens when wages are too high relative to labor productivity. Thus, to reduce "U" during bad times, subsidizing wages is a way to effectively estimulate "L^d" when "a" is relatively low.